

A Single-NUFFT Hyperparameter Gradient for EFGP

1 Gradient of the log marginal likelihood

For a Gaussian process with kernel K_θ and noise variance σ^2 ,

$$\log p(y \mid \theta) = -\frac{1}{2}y^\top \tilde{K}^{-1}y - \frac{1}{2}\log \det \tilde{K} - \frac{N}{2}\log 2\pi, \quad \tilde{K} = K_\theta + \sigma^2 I, \quad (1)$$

and for any hyperparameter θ ,

$$\frac{\partial \log p}{\partial \theta} = \frac{1}{2}(T_2(\theta) - T_1(\theta)), \quad T_1 = \text{tr}(\tilde{K}^{-1} \partial_\theta \tilde{K}), \quad T_2 = y^\top \tilde{K}^{-1} \partial_\theta \tilde{K} \tilde{K}^{-1} y. \quad (2)$$

2 EFGP approximation

Given a shift-invariant kernel with spectral density $\hat{k}_\theta$, pick an equispaced grid $\{\xi_k\}_{k=1}^M$ of spacing h and let

$$D = \text{diag}(\sqrt{h^d \hat{k}_\theta(\xi_k)}), \quad D'_\theta = \text{diag}(h^d \partial_\theta \hat{k}_\theta(\xi_k)), \quad \Phi = FD, \quad (3)$$

where $F \in \mathbb{C}^{N \times M}$ is the non-uniform Fourier matrix. Then $K_\theta \approx \Phi \Phi^* = FD^2 F^*$ and $\partial_\theta K_\theta \approx FD'_\theta F^*$ for every kernel hyper θ (variance σ_f^2 included, since for SE $\partial_{\sigma_f^2} \hat{k} = \hat{k}/\sigma_f^2$ gives $D'_{\sigma_f^2} = D^2/\sigma_f^2$). The noise σ^2 is special: $\partial_{\sigma^2} \tilde{K} = I$, not of $FD'F^*$ form.

Two operators are available in $\mathcal{O}(M \log M)$:

- $T := F^* F$ – an $M \times M$ (multi-level) Toeplitz matrix generated by the vector v_{ker} .
- $A_{\text{mean}} := \sigma^2 I + DTD = \sigma^2 I + \Phi^* \Phi$ – the mean-problem Gram.

3 The trace term T_1

3.1 Kernel hypers: BD' feature-space Hutchinson

By cyclicity,

$$T_1(\theta) = \text{tr}(\tilde{K}^{-1} FD'_\theta F^*) = \text{tr}(B D'_\theta), \quad B := F^* \tilde{K}^{-1} F \in \mathbb{C}^{M \times M}. \quad (4)$$

Woodbury expands B in pure M -space:

$$B = \sigma^{-2}(T - TD A_{\text{mean}}^{-1} DT). \quad (5)$$

Applying BD'_θ to an M -vector costs 2 Toeplitz mat-vecs plus 1 CG solve against A_{mean} ; *no NUFFTs*. Hutchinson with Rademacher probes $v_k \in \{\pm 1\}^M$ gives

$$T_1(\theta) \approx \frac{1}{T_p} \sum_{k=1}^{T_p} v_k^* B D'_\theta v_k. \quad (6)$$

Why this form (and not the “split” form). The equally-valid identity $T_1 = \sigma^{-2} \text{tr}(TD'_\theta) - \sigma^{-2} \text{tr}(A_{\text{mean}}^{-1} DTD'_\theta TD)$ supports Hutchinson on $A_{\text{split}} := \sigma^{-2} A_{\text{mean}}^{-1} DTD'_\theta TD$. It has the same trace but the $DT \cdots TD$ sandwich introduces a factor of $\|F^*F\|^2 \sim N^2$ into $\|A_{\text{split}}\|_F^2$. Empirically this makes the estimator $\sim 10^4 \times$ noisier. The BD' form, by contrast, shares the nonzero eigenvalues of the full $N \times N$ operator $\tilde{K}^{-1}FD'F^*$ and its Frobenius norm matches to within a few percent (see §7).

3.2 Noise σ^2 : existing Woodbury shortcut

For $\partial_{\sigma^2} \tilde{K} = I$, the Woodbury identity $\text{tr}(\tilde{K}^{-1}) = N/\sigma^2 - \sigma^{-2} \text{tr}(A_{\text{mean}}^{-1} \Phi^* \Phi)$ reduces $T_1(\sigma^2)$ to a Hutchinson estimate of a trace over $\Phi^* \Phi = DTD$, again purely in M -space and already in the codebase.

3.3 Variance σ_f^2 : closed-form from the noise estimator

Since K_θ is linear in σ_f^2 for the SE/Matérn family, $\partial_{\sigma_f^2} K = K/\sigma_f^2$, and the variance trace reduces algebraically to the noise trace:

$$T_1(\sigma_f^2) = \frac{1}{\sigma_f^2} \text{tr}(\tilde{K}^{-1}K) = \frac{1}{\sigma_f^2} \left(\text{tr}(I) - \sigma^2 \text{tr}(\tilde{K}^{-1}) \right) = \frac{N - \sigma^2 T_1(\sigma^2)}{\sigma_f^2}. \quad (7)$$

This is the same shortcut already present in the current code (`efgpnd.py:266`); we use it in the proposed path as well, so variance *is not* a separate Hutchinson hyper. Only the kernel hypers $\theta \neq \sigma_f^2$ contribute trace-probe RHSs.

3.4 Single batched CG for the entire trace term

Group all Hutchinson RHSs into one call. Let $\theta_1, \dots, \theta_{K-1}$ be the kernel hypers excluding variance and let $V \in \{\pm 1\}^{T_p \times M}$, $V_n \in \{\pm 1\}^{T_p \times M}$ be the kernel- and noise-probe Rademacher matrices. Stack

$$\text{rhs} = \begin{bmatrix} DT D'_{\theta_1} V \\ \vdots \\ DT D'_{\theta_{K-1}} V \\ DTDV_n \end{bmatrix} \in \mathbb{C}^{((K-1)+1)T_p \times M} \quad (8)$$

and solve $A_{\text{mean}} X = \text{rhs}$ with one batched CG. The post-processing – one Toeplitz mat-vec to recover $BD'_\theta V$ for the kernel block, and an M -dot to extract $\text{tr}(K_{\text{tilde}}^{-1})$ from the noise block – is $\mathcal{O}(M \log M)$. **Total: 2 CG calls (mean + trace)**, matching the current implementation exactly.

4 The quadratic form T_2 : current vs. proposed

Write $\alpha := \tilde{K}^{-1}y$ and $\beta_0 := A_{\text{mean}}^{-1}DF^*y$ for the mean-problem solution. Woodbury gives

$$\alpha = \sigma^{-2}(y - \Phi\beta_0) = \sigma^{-2}(y - FD\beta_0), \quad F^*\alpha = \sigma^{-2}(F^*y - T(D\beta_0)). \quad (9)$$

Crucially, $F^*\alpha$ is computed via a single *Toeplitz* mat-vec, not an NUFFT.

4.1 Current implementation (`efgpnd.py`)

The variance shortcut and noise line both read α as an N -vector, which is why line 3’s NUFFT exists.

Algorithm 1 Current T_2 path for all hypers.

- | | |
|---|--|
| 1: $Fy \leftarrow \text{NUFFT}^*(y)$ | ▷ NUFFT #1 (setup) |
| 2: Solve $A_{\text{mean}}\beta_0 = D \cdot Fy$ by CG. | |
| 3: $z \leftarrow \text{NUFFT}(D\beta_0)$ | ▷ NUFFT #2 – forms α in N -space |
| 4: $\alpha \leftarrow (y - z)/\sigma^2$ | |
| 5: $F^*\alpha \leftarrow (Fy - T(D\beta_0))/\sigma^2$ | ▷ Toeplitz, no NUFFT |
| 6: For each kernel hyper $\theta \neq$ variance: $T_2(\theta) = \langle F^*\alpha, D'_\theta F^*\alpha \rangle$ | ▷ M -space, generic path |
| 7: Variance (shortcut): $T_2(\sigma_f^2) = (y^\top \alpha - \sigma^2 \ \alpha\ ^2)/\sigma_f^2$ | ▷ uses N-space α |
| 8: Noise: $T_2(\sigma^2) = \ \alpha\ ^2$ | ▷ uses N-space α |
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4.2 Proposed implementation (single NUFFT)

Two observations eliminate line 3:

(a) Variance has a generic D' path – which blows up under loose CG. Since $\partial_{\sigma_f^2} \hat{k} = \hat{k}/\sigma_f^2$ gives $D'_{\sigma_f^2} = D^2/\sigma_f^2$,

$$T_2(\sigma_f^2) = \langle F^*\alpha, D'_{\sigma_f^2} F^*\alpha \rangle = \|D F^*\alpha\|^2/\sigma_f^2 \quad (10)$$

is mathematically exact. Numerically, however, $F^*\alpha$ is built via the Toeplitz residual $F^*\alpha = (Fy - T(D\beta_0))/\sigma^2$, and at inexact CG the quantity $D F^*\alpha$ picks up the CG residual amplified by $1/\sigma^2$. Writing $r = DFy - A_{\text{mean}}\beta_0$,

$$D F^*\alpha = (DFy - DTD\beta_0)/\sigma^2 = \beta_0 + r/\sigma^2, \quad (11)$$

so $\|D F^*\alpha\|^2 = \|\beta_0\|^2 + 2\Re\{\beta_0^* r\}/\sigma^2 + \|r\|^2/\sigma^4$. The $\|r\|^2/\sigma^4$ term is *quadratic* in the residual and wrecks the gradient when σ^2 shrinks during training (e.g. $\sigma^2 \sim 10^{-2}$, $\text{cg_tol} = 10^{-3}$, $N = 2.5 \cdot 10^5$: the error flips the sign of $\nabla_{\sigma_f^2}$ and drives σ_f^2 to infinity). The fix: use

$T_2(\sigma_f^2) \sigma_f^2 = \|\beta_0\|^2,$

(12)

which is the same value at CG convergence ($DF^*\alpha = \beta_0$), but whose error is *linear* in the CG residual ($O(\text{cg_tol})$, no $1/\sigma^2$ or $1/\sigma^4$ amplification). All other kernel hypers stay on the D' path.

(b) Noise needs a scalar Woodbury identity. Since $\partial_{\sigma^2} \tilde{K} = I$ is *not* of $FD'F^*$ form, there is no sandwich trick for the noise T_2 . The scalar $\|\alpha\|^2$ has a direct M -space expansion:

$\|\alpha\|^2 = \sigma^{-4}(\|y\|^2 - 2\Re\{\beta_0^* DF^*y\} + \beta_0^* DTD\beta_0).$

(13)

This is the literal expansion of $\|(y - FD\beta_0)/\sigma^2\|^2$ and does *not* invoke $A_{\text{mean}}\beta_0 = DF^*y$, so it is robust to CG residual (unlike the Woodbury-simplified $\sigma^{-4}(\|y\|^2 - c - \sigma^2\|\beta_0\|^2)$, which leaks $\beta_0^* r/\sigma^4$). All ingredients are on hand after the CG solve: $c := \Re\{\beta_0^* DFy\}$ is needed anyway, and $\beta_0^* DTD\beta_0 = \langle \beta, T\beta \rangle$ with $\beta := D\beta_0$ reuses the already-computed $T\beta$ from $F^*\alpha$. Cost: $\mathcal{O}(M)$. Everything after the single setup NUFFT F^*y lives in M -space.

Algorithm 2 Proposed T_2 path for all hypers.

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|--|------------------------------------|
| 1: $Fy \leftarrow \text{NUFFT}^*(y)$ | ▷ NUFFT #1 – the only NUFFT |
| 2: $\ y\ ^2 \leftarrow \langle y, y \rangle$ | ▷ $\mathcal{O}(N)$ setup scalar |
| 3: Solve $A_{\text{mean}}\beta_0 = D \cdot Fy$ by CG. | |
| 4: $F^*\alpha \leftarrow (Fy - T(D\beta_0))/\sigma^2$ | ▷ Toeplitz |
| 5: $c \leftarrow \Re\{\beta_0^* D Fy\}$ | ▷ reusable M -dot |
| 6: For each kernel hyper $\theta \neq \sigma_f^2$: $T_2(\theta) = \langle F^*\alpha, D'_\theta F^*\alpha \rangle$ | ▷ M -space |
| 7: Variance (CG-robust): $T_2(\sigma_f^2) = \ \beta_0\ ^2/\sigma_f^2$ | ▷ M -space scalar |
| 8: Noise: $T_2(\sigma^2) = \sigma^{-4}(\ y\ ^2 - 2c + \beta_0^* D T D \beta_0)$ | ▷ M -space scalar, CG-robust |
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Can we drop the NUFFT for noise too? Yes. The forward NUFFT in the current code exists solely to materialize α for $\|\alpha\|^2$ (and $y^\top \alpha$ in the variance shortcut and in the log-marginal). With the scalar Woodbury identity above, $\|\alpha\|^2$ is available with no NUFFT. Even though the noise-derivative is the identity and thus cannot use the $FD'F^*$ sandwich, its scalar *can* be expressed entirely in M -space via the same substitution that gives $F^*\alpha$ – Woodbury applied to the *vector* α rather than to an operator. So the only truly irreducible NUFFT in the whole gradient is the setup F^*y .

5 Total NUFFT and CG count

Let K_{tr} be the number of *trace* kernel hypers – kernel hypers excluding variance, since variance uses the closed-form shortcut of §3.3. (For SE in any d with a single lengthscale, $K_{\text{tr}} = 1$; for ARD with d lengthscales, $K_{\text{tr}} = d$.) Let T_p be the number of Hutchinson probes.

Stage	Current	Proposed
Term 2 setup F^*y	1	1
Term 2 α -materialization $F(D\beta_0)$	1	0
Term 1 trace probes F^*Z (batched)	1	0
Term 1 $F(D'_\theta F^*Z)$ (batched)	1	0
Term 1 $F(D\beta_{\text{trace}})$ (batched)	1	0
Total NUFFT calls	5	1
CG calls	2	2
Trace CG batch size	$(K_{\text{tr}} + 1) T_p$	$(K_{\text{tr}} + 1) T_p$

Table 1: NUFFT and CG accounting. The proposed path matches the current path’s CG structure exactly – one mean solve plus one batched trace solve over $(K_{\text{tr}}+1)T_p$ RHSs – and removes 4 of the 5 NUFFT calls. Each “NUFFT call” is a single batched transform; the work per call scales with the batch (e.g., T_p for the trace probes).

Per-hyper cost for Term 1 after the switch is $T_p \times (2 \text{ Toeplitz} + 1 \text{ CG})$, all $\mathcal{O}(M \log M)$; per-hyper cost for Term 2 is $\mathcal{O}(M)$ (one vector product).

Algorithm 3 Full log-marginal gradient with a single NUFFT and 2 CG calls.

- 1: **Setup:** build v_{ker} for T ; form D , all $\{D'_\theta\}$ via `spectral_grad`; form Jacobi preconditioner for A_{mean} .
 - 2: $Fy \leftarrow \text{NUFFT}^*(y)$; $\|y\|^2 \leftarrow \langle y, y \rangle$ (only NUFFT, $\mathcal{O}(N + M \log M)$)
 - 3: Solve $A_{\text{mean}}\beta_0 = D \cdot Fy$ by CG. (CG #1)
 - 4: $F^*\alpha \leftarrow (Fy - T(D\beta_0))/\sigma^2$ ($M \log M$)
 - 5: $c \leftarrow \Re\{\beta_0^* D F y\}$
 - 6: **Term 2 – non-variance kernel hypers:** $T_2(\theta) \leftarrow \langle F^*\alpha, D'_\theta F^*\alpha \rangle$
 - 7: **Term 2 – variance (CG-robust):** $T_2(\sigma_f^2) \leftarrow \|\beta_0\|^2/\sigma_f^2$
 - 8: **Term 2 – noise (CG-robust):** $T_2(\sigma^2) \leftarrow \sigma^{-4}(\|y\|^2 - 2c + \beta_0^* D T D \beta_0)$
 - 9: Draw $V, V_n \in \{\pm 1\}^{T_p \times M}$.
 - 10: **Stack RHSs** for Term 1:
 - 11: kernel block (skip variance): for each trace hyper θ , $u_\theta \leftarrow D'_\theta V$; $y_1^\theta \leftarrow T u_\theta$; $\text{rhs}^\theta \leftarrow D y_1^\theta$.
 - 12: noise block: $\text{rhs}^{\sigma^2} \leftarrow D T(D V_n)$.
 - 13: Solve one batched $A_{\text{mean}}X = \text{stacked rhs}$ by CG. (CG #2)
 - 14: **Term 1 – kernel hypers:** for each θ , $y_2^\theta \leftarrow T(DX^\theta)$, $T_1(\theta) \leftarrow \text{mean}_k \langle V_k, (y_1^\theta - y_2^\theta)/\sigma^2 \rangle$.
 - 15: **Term 1 – noise:** $T_1(\sigma^2) \leftarrow N/\sigma^2 - \sigma^{-2} \text{mean}_k \langle V_{n,k}, X_k^{\sigma^2} \rangle$.
 - 16: **Term 1 – variance (closed form, §3.3):** $T_1(\sigma_f^2) \leftarrow (N - \sigma^2 T_1(\sigma^2))/\sigma_f^2$.
 - 17: **Return** $\frac{1}{2}(T_2 - T_1)$ for each hyper.
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6 Algorithm: full gradient, single NUFFT

7 Empirical timing and variance

7.1 Per-probe Hutchinson variance

Compared $T_p = 1$ Hutchinson std vs. dense ground truth, $N_{\text{rep}} = 60$ replications, SE kernel with $\sigma_f^2 = 1, \sigma^2 = 0.1$, CG tol 10^{-8} , EFGP at $\varepsilon = 10^{-4}$.

d	N	ℓ	M	current N -space std	BD' M -space std	ratio
1	4000	0.04	41	1.23×10^3	3.15×10^2	0.26 $\times$
1	4000	0.08	25	5.50×10^2	2.05×10^2	0.37 $\times$
1	4000	0.30	13	1.55×10^2	7.66×10^1	0.49 $\times$
2	2000	0.08	729	1.11×10^3	6.45×10^2	0.58 $\times$
2	2000	0.15	361	4.91×10^2	3.20×10^2	0.65 $\times$
2	5000	0.08	729	1.62×10^3	8.38×10^2	0.52 $\times$

Table 2: Per-probe Hutchinson std for the lengthscale Term-1 trace. BD' is consistently lower-variance than the current N -space form (40–75% reduction), in both $d = 1$ and $d = 2$, across the lengthscale range. The “split” Woodbury form (not shown here) was $\sim 10^4 \times$ noisier and is the wrong M -space identity to Hutchinson against.

7.2 Single-NUFFT variance comparison (held out for reference)

Original $d = 1, N = 4000, \ell = 0.08, M = 31, \varepsilon = 10^{-6}$ comparison including all three estimators:

Trace estimator	$\ A\ _F^2$	theoretical std/probe	empirical std/probe
$A_N = \tilde{K}^{-1}FD'F^*$ (current N -space)	1.97×10^5	626	545
$A_{\text{split}} = \sigma^{-2}A_{\text{mean}}^{-1}DTD'TD$	2.62×10^{10}	127,094	63,179
BD' (proposed)	2.04×10^5	511	250

Table 3: Why the right Woodbury form matters: split has the same trace but $\|\cdot\|_F^2 \sim N^2$, killing the Hutchinson estimator. BD' shares eigenvalues with A_N and is marginally tighter on the off-diagonal mass.

Probes T_p	current N -space	BD' M -space	exact dense (§7.4)
10	5.5 ms	3.1 ms	
50	13.0 ms	4.5 ms	0.6 ms (deterministic, zero variance)
200	46.0 ms	15.8 ms	

Table 4: Wall-clock per trace evaluation at fixed CG tolerance. Combined with the $\sim 5\times$ lower per-probe variance, BD' is roughly 10–15 $\times$ faster than the current path for equal accuracy.

7.3 End-to-end full-gradient timing

After applying both refinements (variance shortcut + single batched CG), we time the entire $\partial_\theta \log p$ evaluation against the current implementation. SE kernel, $\ell = 0.08$ (sweep over $\ell \in \{0.04, 0.08, 0.15, 0.30\}$), $\sigma_f^2 = 1, \sigma^2 = 0.1$, EFGP at $\varepsilon = 10^{-4}$, CG tol 10^{-5} , $T_p = 1$, NUFFT eps 6×10^{-8} . Times are means of 3 reps after warmup.

d	N	M	current (ms)	proposed (ms)	speedup
1	10,000	25	19.6	17.2	1.14 $\times$
1	50,000	25	22.6	18.4	1.23 $\times$
1	100,000	25	23.8	18.7	1.27 $\times$
1	500,000	25	39.3	23.8	1.65 $\times$
2	10,000	729	83.4	67.1	1.24 $\times$
2	50,000	729	100.8	85.8	1.18 $\times$
2	100,000	729	103.7	107.2	0.97 $\times^*$
2	500,000	729	163.0	120.3	1.35 $\times$

Table 5: End-to-end gradient timing at $T_p = 1, \ell = 0.08$. *The single sub-unity row is within timing noise: the same lengthscale at $N = 10\text{k}$ and $N = 500\text{k}$ both win, so this is not a real regression. Across the full sweep ($\ell \in \{0.04, \dots, 0.30\}$, $N \in \{10\text{k}, \dots, 500\text{k}\}$), proposed wins on every other configuration with speedups 1.04 $\times$ to 1.77 $\times$. Speedup grows with N (NUFFT cost dominates) and with larger ℓ (smaller M makes M -space ops cheaper). At higher T_p the win is much larger – e.g. at $d = 1, N = 100\text{k}, T_p = 50$, current = 125 ms vs. proposed = 33 ms (**3.79 $\times$**).

7.4 When M is small: go dense

For $\ell = 0.08$ above, $M = 31$, and forming the $M \times M$ operator explicitly costs $\mathcal{O}(M^3) \approx 30\text{k}$ flops. The deterministic solve

$$T_1(\theta) = \sigma^{-2} \text{tr}(TD'_\theta) - \sigma^{-2} \text{tr}(A_{\text{mean}}^{-1}DTD'_\theta TD), \quad (14)$$

evaluated densely, runs in 0.6 ms, gives 10-digit accuracy, and beats every Hutchinson variant. A full lengthscale sweep (Table 6) shows this holds across $\ell \in [0.02, 0.6]$ with $M \in [13, 95]$.

ℓ	M	time (ms)		error std (at T_p)		
		exact	BD' ($T_p=10$)	current $T_p=1$ / BD'	$T_p=4$	$T_p=10$
0.02	95	2.4	4.3	2400/ 888	1490/ 438	1040/ 296
0.04	53	1.1	3.2	1340/ 506	652/ 229	429/ 167
0.08	31	0.6	3.1	500/ 256	299/ 132	181/ 89
0.15	21	0.4	2.5	278/ 148	143/ 75	86/ 46
0.30	17	0.3	2.5	102/ 112	75/ 64	42/ 46
0.60	13	0.3	2.4	89/ 37	49/ 17	31/ 14

Table 6: Lengthscale sweep. At these values of M the exact dense solve dominates; among Hutchinson variants, BD' strictly beats the current N -space path. Crossover to favoring stochastic estimation occurs around $M \gtrsim 300$ –1000, which for $d = 1$ SE/Matern rarely happens but is typical in $d = 2, 3$.

8 Summary

- Cyclicity + Woodbury collapse the trace T_1 to $\text{tr}(BD'_\theta)$ in M -space, estimated with $\mathcal{O}(T_p M \log M)$ work and *no* NUFFTs.
- The quadratic form T_2 is already nearly M -space after the setup NUFFT F^*y . The current code’s forward NUFFT to form α is used only by the variance shortcut, the noise scalar, and the log-marginal. Replacing the variance shortcut with the generic D' path and the noise scalar with the M -space Woodbury identity removes that NUFFT too.
- **Every kernel hyper, including variance, uses the same D' -sandwich form for T_2 ; only the noise σ^2 needs a dedicated scalar identity, and that identity is also pure M -space.**
- For T_1 , variance has the same closed-form shortcut $T_1(\sigma_f^2) = (N - \sigma^2 T_1(\sigma^2))/\sigma_f^2$ used by the current code, so it is *not* a separate Hutchinson hyper. The kernel-trace and noise-trace RHSs stack into one batched CG, matching the current code’s CG count exactly.
- **Total NUFFT count across the full gradient drops from 5 to 1; CG count is unchanged at 2.**
- Per-probe Hutchinson variance for BD' is consistently 0.26–0.65 $\times$ the current N -space variance across $d \in \{1, 2\}$ and $\ell \in [0.04, 0.30]$ – the proposed estimator is always tighter.
- When M is small (1D SE/Matern, coarse 2D), replace Hutchinson by a dense $\mathcal{O}(M^3)$ solve for exact, deterministic traces.

9 What’s left to improve?

The proposed pipeline now hits the structural minimum on the two operations whose cost actually scales with N : one NUFFT F^*y and an $\mathcal{O}(N)$ inner product $\|y\|^2$. Everything else is $\mathcal{O}(M \log M)$ Toeplitz/CG work. Concretely:

- **Cannot remove the lone NUFFT.** F^*y is the only data-dependent ingredient; without it neither T_1 nor T_2 has access to y in any form (all subsequent operations factor through Fy and $\|y\|^2$).
- **Cannot reduce CG count.** The mean solve gives β_0 (used in both T_1 via A_{mean} structure and T_2 via $F^*\alpha$); the trace solve estimates $\text{tr}(BD')$. These cannot share a single CG because their RHSs differ.
- **Marginal opportunities:**
 - Sharing CG warm-starts across optimizer iterations (already supported via `mean.cg.init`).
 - Reusing the same V probes across optimizer iterations (variance reduction via correlated probes).
 - Block preconditioner that captures more of A_{mean} 's structure than Jacobi – only useful when CG is the bottleneck (large M , $d \geq 2$).
 - For small M (1D, coarse 2D), the dense $M \times M$ solve already wins by orders of magnitude (§7.4); the codebase could auto-route to dense when $M \lesssim 200$.